

Chapter II Exercise 2.9–2.19

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July 30, 2026

2.9. If X is a topological space, and Z an irreducible closed subset of X , a *generic point* for Z is a point ζ such that $Z = \{\zeta\}^-$. If X is a scheme, show that every (nonempty) irreducible closed subset has a unique generic point.

Solution. The intersection of Z with every (non-empty) affine open U in X has a unique generic point given by taking the radical ideal (here we applied the assumption that Z , and thus $Z \cap U$, is irreducible). Any two open subsets of Z intersect, so the generic points are actually the same point in X (using the uniqueness in the above). $\square$

2.10. Describe $\text{Spec } \mathbb{R}[x]$. How does its topological space compare to the set $\mathbb{R}$? To $\mathbb{C}$?

Solution. I don't fully understand the question, but I will try. $\mathbb{R}[X]$ is a PID, so all non-zero prime ideals are maximal and simply correspond to the monic irreducible polynomials. Hence there are 3 types of points, classified by their residue fields:

- The generic point η , which has $\kappa(\eta) = \mathbb{R}(X)$;
- Closed points $(X - \alpha)$ where $\alpha \in \mathbb{R}$, having $\kappa = \mathbb{R}$;
- Closed points corresponding to pairs of conjugate complex numbers $(X^2 + aX + b)$ where $a^2 - 4b < 0$. These have $\kappa = \mathbb{C}$.

On the closed points, this is almost just the Zariski topology on $\mathbb{C}$, except that we can't distinguish conjugate pairs of numbers. In particular, it restricts to the usual Zariski topology on $\mathbb{R}$. $\square$

2.11. Let $k = \mathbf{F}_p$ be the finite field with p elements. Describe $\text{Spec } k[x]$. What are the residue fields of its points? How many points are there with a given residue field?

Solution. As above, this has a generic point η with $\kappa(\eta) = k(X)$, and its closed points correspond to monic irreducible polynomials. If A is a ring, the residue field of $\text{Spec}(A)$ at prime $\mathfrak{p}$ is the field of fractions of $A/\mathfrak{p}$. (Clearly equivalent to usual definition as residue field of local ring.) Of course, for our closed points, $A/\mathfrak{p}$ is already a field. Its dimension over k is given by the degree of the generator of $\mathfrak{p}$, so $\kappa(\mathfrak{p}) \simeq \mathbb{F}_{p^d}$.

Hence counting points with a given residue field is the same as counting number of monic irreducible polynomials of a given degree. This has a classical solution using Möbius inversion formula, see <https://math.stackexchange.com/questions/92747/counting-irreducible-polynomials>. $\square$

2.13. A topological space is *quasi-compact* if every open cover has a finite subcover.

- (a) Show that a topological space is noetherian (I, §1) if and only if every open subset is quasi-compact.
- (b) If X is an affine scheme, show that $\text{sp}(X)$ is quasi-compact, but not in general noetherian. We say a scheme X is *quasi-compact* if $\text{sp}(X)$ is.
- (c) If A is a noetherian ring, show that $\text{sp}(\text{Spec } A)$ is a noetherian topological space.
- (d) Give an example to show that $\text{sp}(\text{Spec } A)$ can be noetherian even when A is not.

Solution. (a) $\implies$: In fact, if X is Noetherian, then any subset $H \subset X$ is quasi-compact. Suppose $\{F_i\}_{i \in I}$ is a filtered family of closed subsets of H , and $\bigcap_{i \in I} F_i = \emptyset$, we need to show that some $F_i = \emptyset$. By the Noetherian property, the filtered family $\{\text{cl}_X(F_i)\}_{i \in I}$ stabilizes. Since $\text{cl}_X(F_i) \cap H = \text{cl}_H(F_i) = F_i$, the original family $\{F_i\}_{i \in I}$ stabilizes as well.

$\impliedby$: If $Z_1 \supset Z_2 \supset \dots$ is a descending chain of closed subsets, then the complements Z_n^c gives an open cover of $(\bigcap_n Z_n)^c$. By quasi-compactness we can extract a finite subcover, and the result follows.

- (b) In fact, every distinguished open is quasi-compact. This was shown in lecture.
- (c) Obvious.

(d) Let k be a field. Consider $A = \bigcup_{t \in \mathbb{R}_{\geq 0}} k[[x^t]]$. Being a valuation ring, its ideals falls into either of the following 3 classes:

- (0) .
- (x^t) for some $t \geq 0$. In other words this is $\{a \in A : v(a) \geq t\}$.
- $(x^{>t}) := \bigcup_{s > t} (x^s)$ for some $t \geq 0$. In other words this is $\{a \in A : v(a) > t\}$.

Here v denotes the valuation.

A is clearly not Noetherian, since

$$(x) \subset (x^{1/2}) \subset (x^{1/3}) \subset \dots$$

does not stabilize.

However, A only has two prime ideals: (0) and $(x^{>0})$, so $\text{sp}(\text{Spec } A)$ is trivially a Noetherian space. $\square$

- 2.14.** (a) Let S be a graded ring. Show that $\text{Proj } S = \emptyset$ if and only if every element of S_+ is nilpotent.
- (b) Let $\varphi: S \rightarrow T$ be a graded homomorphism of graded rings (preserving degrees). Let $U = \{\mathfrak{p} \in \text{Proj } T \mid \mathfrak{p} \not\supseteq \varphi(S_+)\}$. Show that U is an open subset of $\text{Proj } T$, and show that φ determines a natural morphism $f: U \rightarrow \text{Proj } S$.
- (c) The morphism f can be an isomorphism even when φ is not. For example, suppose that $\varphi_d: S_d \rightarrow T_d$ is an isomorphism for all $d \geq d_0$, where d_0 is an integer. Then show that $U = \text{Proj } T$ and the morphism $f: \text{Proj } T \rightarrow \text{Proj } S$ is an isomorphism.
- (d) Let V be a projective variety with homogeneous coordinate ring $S(\mathbb{I}, \S 2)$. Show that $t(V) \cong \text{Proj } S$.

Remark. (b) expresses that Proj is not a functor, and (c) shows that it loses information about the graded ring (in comparison, Spec is fully faithful functor).

Solution. (a) If an element $a = a_0 + a_1 + a_2 + \dots$ of S is nilpotent, where $a_d \in S_d$, then a_0 is nilpotent since it's the lowest degree term. Inductively, this shows that every a_d is

nilpotent. Hence the nilradical $\text{nil } S$ of S is a homogeneous ideal. It is the intersection of all prime ideals of S .

I claim that it is also the intersection of all homogeneous prime ideals of S . Then we win.

We just need that every prime contains a minimal prime. Fix a prime ideal $\mathfrak{p}$. Let $\mathfrak{q}$ be the homogeneous ideal generated by all homogeneous elements of $\mathfrak{p}$, then $\mathfrak{q} \subset \mathfrak{p}$. I claim that this $\mathfrak{q}$ is prime. If $c = ab \in \mathfrak{q}$, we can write

$$c = \sum_{d \geq 0} c_d, \quad c_0 = a_0 b_0$$

Since $\mathfrak{q}$ is a homogeneous ideal, $c_0 \in \mathfrak{q} \subset \mathfrak{p}$. By primeness, either $a_0 \in \mathfrak{p}$ or $b_0 \in \mathfrak{p}$; say $a_0 \in \mathfrak{p}$. But since a_0 is homogeneous of degree 0, we actually get $a_0 \in \mathfrak{q}$. So we can subtract a_0 from a , and start looking at $c_1 = a_1 b_0$. Continuing in this fashion, we can show that either all a_d 's or all b_d 's are in $\mathfrak{q}$. Thus $\mathfrak{q}$ is prime. $\square$

Side note. From the proof, we observe the following:

Corollary 0.1. A minimal prime ideal in a graded ring is homogeneous.

Solution (continued). (b) Since everything in $\varphi(S_+)$ is generated by homogeneous elements, $\mathfrak{p} \supset \varphi(S_+)$ iff it contains the homogeneous ideal generated by $\varphi(S_+)$. This is by definition a closed subset, so its complement U is open.

Pulling back along φ gives us the map $f : U \rightarrow \text{Proj } S$. The condition $\mathfrak{p} \not\supset \varphi(S_+)$ guarantees it is well-defined.

We check that this is a scheme morphism on affine charts. Fix a homogeneous $\alpha \in S_+$. Then $\text{Proj}(S)|_{D_+(\alpha)} \simeq \text{Spec } S_\alpha$, and $\text{Proj}(T)$ restricted to U restricted to $D_+(\varphi(\alpha)) \subset U$ is isomorphic to $\text{Spec } T_{\varphi(\alpha)}$. So on these charts the scheme morphism is the ring homomorphism induced by the universal property of localization. These are obviously compatible on intersections, so glue into a global scheme morphism f .

(c) If $\varphi_d : S_d \rightarrow T_d$ are isomorphisms past a degree d_0 , then any prime ideal containing $\varphi(S_+)$ will contain $\bigoplus_{d \geq d_0} T_d$, and thus contain the entire T_+ . Hence

$$U = \{\mathfrak{p} \in \text{Proj } T : \mathfrak{p} \not\supset \varphi(S_+)\} = \text{Proj } T.$$

We now need to show that $f : \text{Proj } T \rightarrow \text{Proj } S$, $\mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$ is an isomorphism.

First we show that f is injective on points. Suppose $\varphi^{-1}(\mathfrak{p}) = \varphi^{-1}(\mathfrak{q})$. By assumption, $\mathfrak{q} \cap T_{>d_0} = \mathfrak{p} \cap T_{>d_0}$. Replacing $\mathfrak{q}$ with $\mathfrak{p} \cap \mathfrak{q}$, we may assume $\mathfrak{q} \subset \mathfrak{p}$. To get a contradiction,

let $a \in \mathfrak{p} \setminus \mathfrak{q}$ be a homogeneous element. If $\deg(a) > 0$, we get a contradiction by considering $a^{d_0} \in \mathfrak{p} \cap T_{>d_0} \subset \mathfrak{q}$. So it must be that $\mathfrak{q} \cap T_+ = \mathfrak{p} \cap T_+$ and $\deg(a) = 0$. However, for any element $b \in T_+$, we have $ab \in \mathfrak{p} \cap T_+ \subset \mathfrak{q}$. But $a \notin \mathfrak{q}$, so $T_+ \subset \mathfrak{q}$. This is a contradiction with $\mathfrak{q} \in \text{Spec } T$. We conclude that f is injective on points.

Next, we establish that f is surjective and locally an isomorphism. This is enough to guarantee that f is an isomorphism. By Proposition 2.5^[GTM52], we know

$$\text{Proj } S = \bigcup_{\substack{\alpha \in S_d \\ d > 0}} D_+(\alpha)$$

and $D_+(\alpha) \simeq \text{Spec } S_{(\alpha)}$, where $S_{(\alpha)}$ is the subring of degree 0 elements of the localization S_α . Thus we can write

$$f : \text{Proj } T = \bigcup_{\substack{\beta \in T_d \\ d > 0}} D_+(\beta) \longrightarrow \text{Proj } S = \bigcup_{\substack{\alpha \in S_d \\ d > 0}} D_+(\alpha)$$

It suffices to show that, for any α as above, we have an isomorphism

$$f_\alpha : D_+(\varphi(\alpha)) \simeq \text{Spec } T_{(\varphi(\alpha))} \longrightarrow D_+(\alpha) \simeq \text{Spec } S_{(\alpha)}.$$

This is equivalent to checking the following ring homomorphism induced from φ ,

$$\overline{\varphi} : S_{(\alpha)} \longrightarrow T_{(\varphi(\alpha))}$$

is an isomorphism (=bijective).

This is easy. To see surjectivity, if $t/\varphi(\alpha)^n \in T_{(\varphi(\alpha))}$, we have

$$\frac{t}{\varphi(\alpha)^n} = \frac{t\varphi(\alpha)^{d_0}}{\varphi(\alpha)^{n+d_0}} \in \text{im } \overline{\varphi}.$$

To see injectivity, suppose $\overline{\varphi}$ maps u/v and u'/v' to the same element. This means that there is some m , which we may assume to be $\geq d_0$, such that

$$\varphi(\alpha)^m \varphi(u) \varphi(v') = \varphi(\alpha)^m \varphi(u') \varphi(v).$$

Thus $\alpha^m uv' = \alpha^m u'v$, which shows injectivity. This completes the proof.

(d) Glue from affine varieties. On the variety side, we have

$$V = \bigcup_{i=0}^n V \cap U_i, \quad U_i := \{[x_0 : \cdots : x_n] \in \mathbb{P}^n : x_i \neq 0\},$$

where $V \subset \mathbb{P}^n$. Each $V \cap U_i$ is isomorphic to the affine variety, whose coordinate ring is given by $S_{(x_i)}$. (Regular functions on U_i are the polynomials in the symbols x_j/x_i where $j \neq i$).

On the scheme side, we also have

$$\mathrm{Proj} S = \bigcup_{i=0}^n D_+(x_i)$$

and $D_+(x_i) \simeq \mathrm{Spec} S_{(x_i)}$. With enough patience, we can check that the isomorphisms agree on the intersections, but it should be trivial. $\square$

- 2.15.** (a) Let V be a variety over the algebraically closed field k . Show that a point $P \in t(V)$ is a closed point if and only if its residue field is k .
 (b) If $f: X \rightarrow Y$ is a morphism of schemes over k , and if $P \in X$ is a point with residue field k , then $f(P) \in Y$ also has residue field k .
 (c) Now show that if V, W are any two varieties over k , then the natural map

$$\mathrm{Hom}_{\mathbf{Var}}(V, W) \rightarrow \mathrm{Hom}_{\mathbf{Set}/k}(t(V), t(W))$$

is bijective. (Injectivity is easy. **The hard part is to show it is surjective.**)

Note. This is part of proving Proposition 2.6.

Solution. (a) The $\implies$ direction follows from the Nullstellensatz, which says that if $I \subsetneq k[x_1, \dots, x_n]$ is a proper ideal, then it has a zero, so maximal ideals (closed points) must correspond to points in the variety.

Now we show the contrapositive of $\impliedby$. Since closedness is local, the subset $\{P\}$ is closed iff it's closed in every element of an open cover of V by affine charts, we can restrict to the case that V is affine. So assume P corresponds to an affine variety Y with coordinate ring A . We have

$$\mathrm{tr. deg}(\kappa(P)|k) \stackrel{\text{Thm 1.1.8}}{=} \dim A \stackrel{\text{Prop 1.1.7}}{=} \dim Y$$

where $\dim A$ denotes the Krull dimension. It follows that

$$\kappa(P) = k \iff \mathrm{tr. deg}(\kappa(P)|k) = 0 \iff \dim Y = 0 \iff P \text{ is a closed point,}$$

as k is algebraically closed.

(b) We have a commutative diagram of fields

$$\begin{array}{ccc} \kappa(P) & \xleftarrow{\quad} & \kappa(f(P)) \\ & \nwarrow \quad \nearrow & \\ & k & \end{array}$$

The result follows.

(c) We first explain how the *functor* t is built; this has not been completed in the textbook. Let $\psi : V \rightarrow W$ be a morphism of varieties. We define $\varphi : t(V) \rightarrow t(W)$ as $\varphi(Z) = \text{cl}_W(Z)$ on points. This is well-defined, since $\text{cl}_W(Z)$ is closed in W and clearly irreducible. Recall that $t(V)$ is a scheme with structure sheaf $\alpha_*\mathcal{O}_V$, where $\alpha : V \rightarrow t(V)$ is the inclusion map. We have a commutative diagram

$$\begin{array}{ccc} V & \xrightarrow{\psi} & W \\ \downarrow \alpha & & \downarrow \alpha \\ t(V) & \xrightarrow{\varphi} & t(W) \end{array}$$

We now see that the variety morphism ψ naturally induces a scheme morphism $t(V) \rightarrow t(W)$, via

$$\mathcal{O}_{t(W)} = \alpha_*\mathcal{O}_W \xrightarrow{\alpha_*\psi^\#} \alpha_*\psi_*\mathcal{O}_V = \varphi_*\alpha_*\mathcal{O}_V = \varphi_*\mathcal{O}_{t(V)}.$$

We now show that the claimed map between hom sets is surjective. Let $f : t(V) \rightarrow t(W)$ be a morphism of schemes over k . Combining (a) and (b), we see that f maps closed points to closed points. So this induces a set map $g : V \rightarrow W$. We need to show that this is a morphism of varieties, and it maps to f under t .

We can cover $t(W) = \bigcup_i Y_i$ by all of its affine opens $\{Y_i\}$, and cover each $\varphi^{-1}(Y_i)$ by all the affine opens of $t(V)$ contained in it. Note that the structure sheaf of $t(V)$ fully determines $\mathcal{O}_V$, and these affine opens are also affine opens in the corresponding varieties.

Thus we can assume that V and W are affine varieties corresponding to the prime ideals $\mathfrak{p} \subset k[x_1, \dots, x_n]$ and $\mathfrak{q} \subset k[x_1, \dots, x_m]$. For simplicity, write $A = k[x_1, \dots, x_n]/\mathfrak{p}$ and $B = k[x_1, \dots, x_m]/\mathfrak{q}$. We have $V = \text{Spec } A$ and $W = \text{Spec } B$.

By exercise II.2.4, the scheme morphism $\varphi : t(V) \rightarrow t(W)$ induces a unique ring homomorphism $B \rightarrow A$. By Proposition I.3.5, this ring homomorphism induces a unique morphism $\psi : V \rightarrow W$ of varieties.

Now we check that the family $\{\psi_i : V_i \rightarrow W_i\}$ of morphisms between affine varieties agree on intersections. I claim it suffices to prove that, if $V' \subset V$ and $W' \subset W$, where $W' = W \setminus Z(f)$ and $V' = \psi_V^{-1}(W') = V \setminus Z(\psi^*f)$, are non-empty distinguished opens, $f \in k[x_1, \dots, x_m]$, then the square

$$\begin{array}{ccc} V' & \xrightarrow{\psi_{V'}} & W' \\ \downarrow & & \downarrow \\ V & \xrightarrow{\psi_V} & W \end{array} \quad (1)$$

commutes. Indeed, we know V' and W' are also affine. This is due to the well-known fact

$$\left(\frac{k[x_1, \dots, x_m]}{\mathfrak{q}} \right)_f \simeq \frac{k[x_1, \dots, x_m, t]}{\mathfrak{q} + (tf - 1)}.$$

Now, we can take the commutative diagram in $\mathbf{Sch}/k$,

$$\begin{array}{ccc} \mathrm{Spec} A_{\varphi^*f} & \xrightarrow{\varphi|_{t(V')}} & \mathrm{Spec} B_f \\ \downarrow & & \downarrow \\ \mathrm{Spec} A & \xrightarrow{\varphi|_{t(V)}} & \mathrm{Spec} B \end{array}$$

and bring it to $\mathbf{Ring}$ and then $\mathbf{Var}$ through the above procedure. Then we get the commutative diagram (1), as desired.

We conclude that $\psi : V \rightarrow W$ is a valid morphism of varieties. It remains to check $t(\psi) = \varphi$. On points, this is easy to prove on affine opens. The sheaf morphisms of φ and $t(\psi)$ agree on sections over affine opens by construction, which is enough to conclude that they are the same sheaf morphism. This completes the proof. $\square$

- 2.16.** Let X be a scheme, let $f \in \Gamma(X, \mathcal{O}_X)$, and define X_f to be the subset of points $x \in X$ such that the stalk f_x of f at x is not contained in the maximal ideal $\mathfrak{m}_x$ of the local ring $\mathcal{O}_x$.
- (a) If $U = \text{Spec } B$ is an open affine subscheme of X , and if $\bar{f} \in B = \Gamma(U, \mathcal{O}_X|_U)$ is the restriction of f , show that $U \cap X_f = D(\bar{f})$. Conclude that X_f is an open subset of X .
 - (b) Assume that X is quasi-compact. Let $A = \Gamma(X, \mathcal{O}_X)$, and let $a \in A$ be an element whose restriction to X_f is 0. Show that for some $n > 0$, $f^n a = 0$. [Hint: Use an open affine cover of X .]
 - (c) Now assume that X has a finite cover by open affines U_i such that each intersection $U_i \cap U_j$ is quasi-compact. (This hypothesis is satisfied, for example, if $\text{sp}(X)$ is noetherian.) Let $b \in \Gamma(X_f, \mathcal{O}_{X_f})$. Show that for some $n > 0$, $f^n b$ is the restriction of an element of A .
 - (d) With the hypothesis of (c), conclude that $\Gamma(X_f, \mathcal{O}_{X_f}) \cong A_f$.

Solution. (a) Trivial.

(b) Since X is quasi-compact, we can cover it with finitely many affine opens $\{U_i\}$. On each affine open, $a|_{U_i} \in \Gamma(U_i, \mathcal{O}_{U_i})$ is the zero section on the distinguished open $D(\bar{f})$. Write $U_i \simeq \text{Spec } B$. We have $\Gamma(D(\bar{f}), \mathcal{O}_{U_i}) = B_{\bar{f}}$, so there is $n(i) \geq 0$ s.t.

$$\bar{f}^{n(i)} \cdot a|_{U_i} = 0 \text{ in } B = \Gamma(U_i, \mathcal{O}_X).$$

Then we can define $n = \max_i n(i)$. Then $f^n a = 0$ since its restriction to every U_i is 0.

(c) Take a finite cover $\{U_i\}$ by affine opens. On each U_i , $f^{n(i)} b \in \Gamma(U_i \cap X_f, \mathcal{O}_X)$ comes from an element of $\Gamma(U_i, \mathcal{O}_X)$ for sufficiently large $n(i)$ by the definition of localization. Take $n = \max_i n(i)$.

Then, apply (b) to the pairwise intersections $U_i \cap U_j$ and the differences of the sections obtained above (which is 0 on X_f). We see that the difference is annihilated by some $f^{m(i,j)}$. Take $m = \max_{i,j} m(i,j)$.

We conclude that there is a well-defined global section which can be identified with $f^{n+m} b$ on X_f .

(d) Since f is non-vanishing on X_f , the restriction induces a natural ring homomorphism $A_f \rightarrow \Gamma(X_f, \mathcal{O}_X)$. It's injective by (b) and surjective by (c). $\square$

2.17. A Criterion for Affineness.

- (a) Let $f: X \rightarrow Y$ be a morphism of schemes, and suppose that Y can be covered by open subsets U_i , such that for each i , the induced map $f^{-1}(U_i) \rightarrow U_i$ is an isomorphism. Then f is an isomorphism.
- (b) A scheme X is affine if and only if there is a finite set of elements $f_1, \dots, f_r \in A = \Gamma(X, \mathcal{O}_X)$, such that the open subsets X_{f_i} are affine, and $f_1, \dots, f_r$ generate the unit ideal in A . [Hint: Use (Ex. 2.4) and (Ex. 2.16d) above.]

Solution. (a) [Trivial.] It suffices to see f is a homeomorphism, since the sheaf isomorphism is local. Now observe that the properties of being injective, surjective and open are all local on the base.

(b) The direction $\implies$ is trivial so we only prove the converse. The condition says we have a finite affine open cover $\{U_i := X_{f_i}\}$ of X . Observe that each pairwise intersection $U_i \cap U_j$ is a distinguished open $D(f_j) \subset U_i$, so is quasi-compact. Hence the hypothesis of exercise 2.16(d) is satisfied. We thus get a natural isomorphism $\Gamma(U_i, \mathcal{O}_X) \simeq A_{f_i}$.

By exercise 2.4, $\text{Hom}_{\text{Sch}}(X, \text{Spec } A) = \text{Hom}_{\text{Ring}}(A, \Gamma(X, \mathcal{O}))$, so we have a natural choice of comparison morphism $\psi: X \rightarrow \text{Spec } A$ induced by id_A . We wish to show that this ψ is an isomorphism.

We apply (a) to the open cover $\text{Spec } A = \bigcup_i D(f_i) = \bigcup_i \text{Spec } A_{f_i}$. Clearly $\psi^{-1}(D(f_i)) = U_i$. By naturality of exercise 2.4, we have a commutative square

$$\begin{array}{ccc} U_i & \xrightarrow{\sim} & \text{Spec } A_{f_i} \\ \downarrow & & \downarrow \\ X & \xrightarrow{\psi} & \text{Spec } A \end{array}$$

The result follows. □

2.18. In this exercise, we compare some properties of a ring homomorphism to the induced morphism of the spectra of the rings.

- (a) Let A be a ring, $X = \text{Spec } A$, and $f \in A$. Show that f is nilpotent if and only if $D(f)$ is empty.
- (b) Let $\varphi: A \rightarrow B$ be a homomorphism of rings, and let $f: Y = \text{Spec } B \rightarrow X = \text{Spec } A$ be the induced morphism of affine schemes. Show that φ is injective if and only if the map of sheaves $f^\#: \mathcal{O}_X \rightarrow f_*\mathcal{O}_Y$ is injective. Show furthermore in that case f is *dominant*, i.e., $f(Y)$ is dense in X .
- (c) With the same notation, show that if φ is surjective, then f is a homeomorphism of Y onto a closed subset of X , and $f^\#: \mathcal{O}_X \rightarrow f_*\mathcal{O}_Y$ is surjective.
- (d) Prove the converse to (c), namely, if $f: Y \rightarrow X$ is a homeomorphism onto a closed subset, and $f^\#: \mathcal{O}_X \rightarrow f_*\mathcal{O}_Y$ is surjective, then φ is surjective. [Hint: Consider $X' = \text{Spec}(A/\ker \varphi)$ and use (b) and (c).]

Solution. (a) If f is nilpotent then $f \in \mathfrak{p}$ for any prime ideal $\mathfrak{p}$. If f is not nilpotent, then A_f is not the zero ring, so it must have a prime ideal, whose preimage in A is a prime ideal not containing f .

(b) $f^\#$ is injective (mono) iff

$$A_\alpha = \mathcal{O}_X(D(\alpha)) \rightarrow \mathcal{O}_Y(f^{-1}D(\alpha)) = \mathcal{O}_Y(D(\varphi(\alpha))) = B_{\varphi(\alpha)}$$

is injective for every $\alpha \in A$. Taking $\alpha = 1$ this implies φ is injective. Conversely, if φ is injective, then $A_\alpha \rightarrow B_{\varphi(\alpha)}$ is obviously injective as well.

Assuming φ injective, we need to show that every non-empty $D(\alpha)$ contains a prime ideal from B . We have α is not nilpotent. Since φ is injective, $\varphi(\alpha)$ is also not nilpotent. So let's take $\mathfrak{p} \in D(\varphi(\alpha))$ by (a). This means $\varphi(\alpha) \notin \mathfrak{p}$ and so $\alpha \notin \varphi^{-1}(\mathfrak{p})$.

(c) If $\varphi: A \rightarrow B$ is surjective, we know it induces a bijection between $\text{Spec}(B)$ and the set of prime ideals of A that contain $\ker \varphi$. This is clearly a topological embedding $\text{Spec}(B) \rightarrow \text{Spec}(A)$.

Next, we need to surjectivity of $f^\#$ on stalks. By exercise II.1.19,

$$(f_*\mathcal{O}_Y)_x = \begin{cases} \mathcal{O}_{Y,y}, & \text{if } x = f(y), \\ 0, & \text{otherwise.} \end{cases}$$

So, if $x \notin f(Y)$, then $f^\#$ is trivially surjective. Write $y =: \mathfrak{p} \leq B$. We can easily see that $\mathcal{O}_{X,f(y)} = A_{f^{-1}(\mathfrak{p})}$ surjects onto $\mathcal{O}_{Y,y} = B_{\mathfrak{p}}$.

(d) Let $X' = \text{Spec}(A/\ker \varphi)$. φ factorizes as $A \rightarrow A/\ker \varphi \hookrightarrow B$.

Let $g : Y \rightarrow X'$ be the induced morphism. By (b), the map $g^\# : \mathcal{O}_{X'} \rightarrow g_*\mathcal{O}_Y$ is injective (and g is dominant).

Let $h : X' \rightarrow X$ be the induced morphism. By (c), the map $h^\# : \mathcal{O}_X \rightarrow h_*\mathcal{O}_{X'}$ is surjective.

Now, we use $h \circ g = f$. On one hand, surjectivity of $f^\# = h_*g^\# \circ h^\#$ implies surjectivity of $h_*g^\#$. Since h is a closed immersion, this implies $g^\#$ is surjective. (This alone is not enough, since a surjective sheaf morphism does not imply surjectivity on global sections!)

On the other hand, $g(Y) = h^{-1}(f(Y))$ is closed, since by assumption $f(Y)$ is closed. A closed dense subset is the entire space, so g is setwise surjective. The same line applied to closed subsets of Y shows that g is a homeomorphism.

Finally, taking global sections of $g^\# : \mathcal{O}_{X'} \xrightarrow{\sim} g_*\mathcal{O}_Y$ yields $A/\ker \varphi \xrightarrow{\sim} B$ as desired. $\square$

2.19. Let A be a ring. Show that the following conditions are equivalent:

- (i) $\text{Spec } A$ is disconnected;
- (ii) there exist nonzero elements $e_1, e_2 \in A$ such that $e_1e_2 = 0$, $e_1^2 = e_1$, $e_2^2 = e_2$, $e_1 + e_2 = 1$ (these elements are called *orthogonal idempotents*);
- (iii) A is isomorphic to a direct product $A_1 \times A_2$ of two nonzero rings.

Solution. (i) $\implies$ (ii). If $\text{Spec } A = U \sqcup V$ is a disjoint union of open subsets, then $e_1 := 1_U$ and $e_2 := 1_V$ defined regular functions on $\text{Spec } A$.

(ii) $\iff$ (iii) Trivial. We have $A_1 = e_1A$ and $A_2 = e_2A$.

(ii) + (iii) $\implies$ (i). Ideals of $A_1 \times A_2$ decomposes in the same way. An ideal $I \leq A_1 \times A_2$ is prime iff

$$\frac{A_1 \times A_2}{I} = \frac{A_1}{e_1I} \times \frac{A_2}{e_2I}$$

is an integral domain. This is only the case if either $e_1I = A_1$ and e_2I is prime, or $e_2I = A_2$ and e_1I is prime. (The 0 ring is not an integral domain.) So $\text{Spec } A_1 \times A_2 \simeq \text{Spec } A_1 \amalg \text{Spec } A_2$ with the evident structure sheaf. $\square$